Note to other teachers and users of these slides: We would be delighted if you found our material useful for giving your own lectures. Feel free to use these slides verbatim, or to modify them to fit your own needs. If you make use of a significant portion of these slides in your own lecture, please include this message, or a link to our web site: http://cs224w.Stanford.edu

Stanford CS224W: Knowledge Graph Embeddings

CS224W: Machine Learning with Graphs
Charilaos Kanatsoulis and Jure Leskovec, Stanford University

http://cs224w.stanford.edu

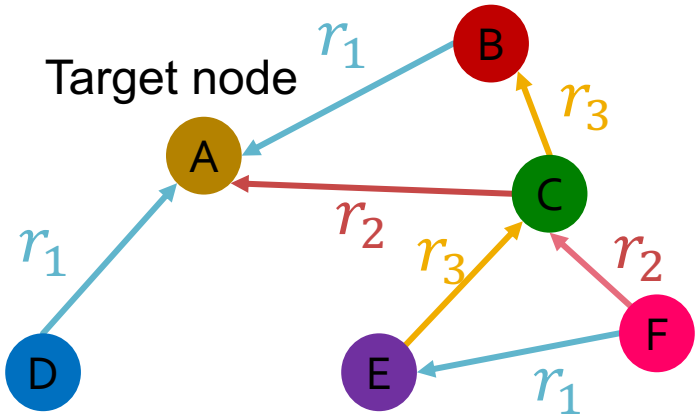


Announcements

- **Colab 2** due **today**
  - Gradescope submissions close at 11:59 PM
- **Colab 3** out **today**, due Thursday 11/5
- **Homework 1 grades are out now!**
  - Regrade requests are open until Thursday 10/29
- **Homework 2** due Thursday 10/29
  - Recitation session materials posted on Ed/Canvas

Recap: Heterogeneous Graphs

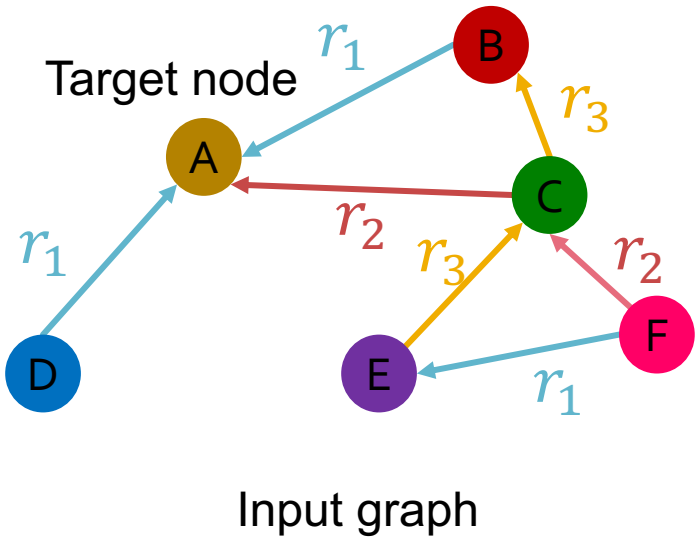
- **Heterogeneous graphs: a graph with multiple relation types**

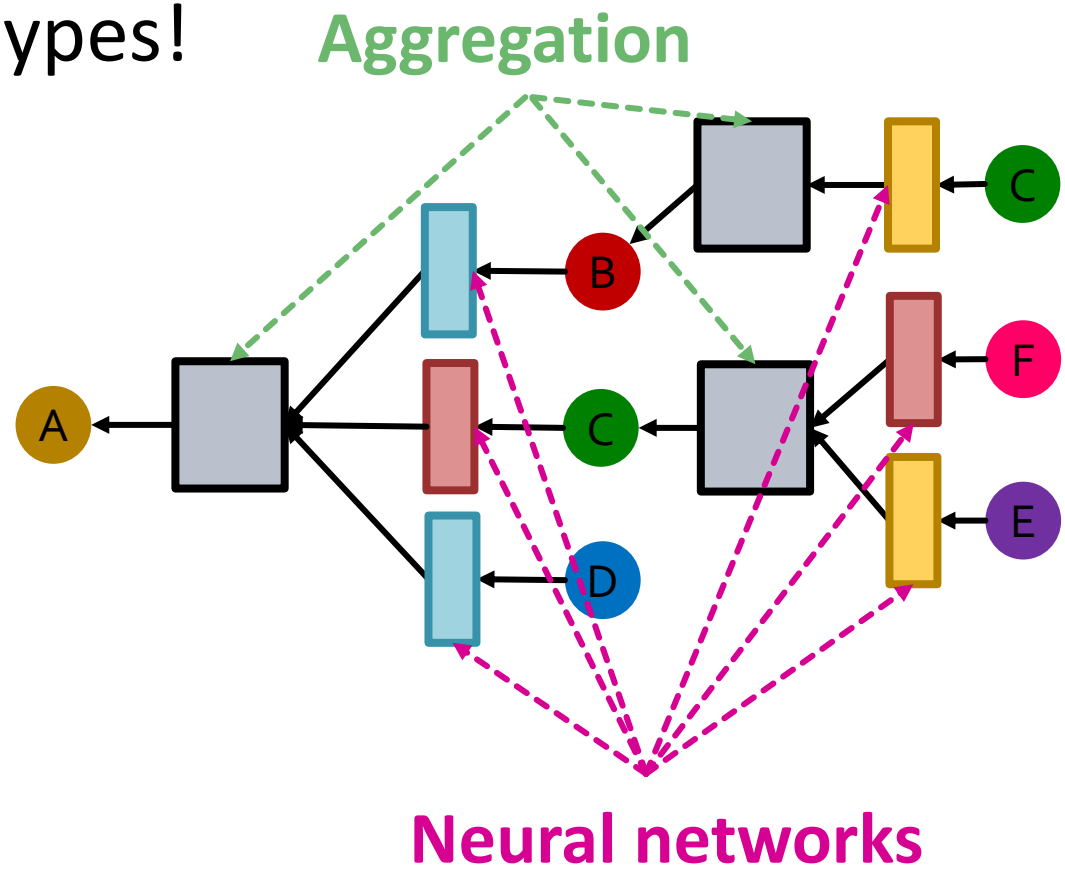


Input graph

Recap: Relational GCN

- Learn from a graph with **multiple relation types**
- Use different neural network weights for different relation types!

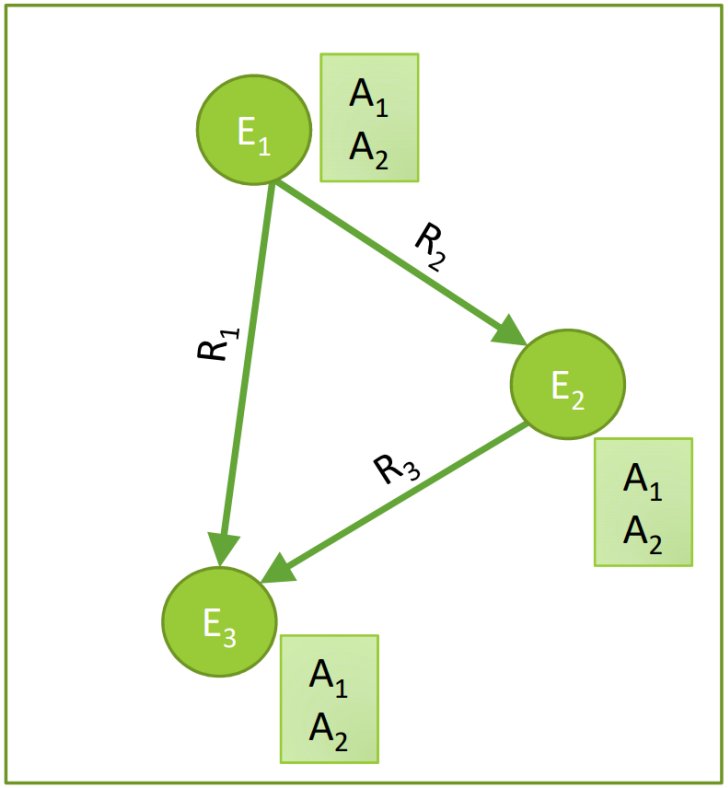




Today: Knowledge Graphs (KG)

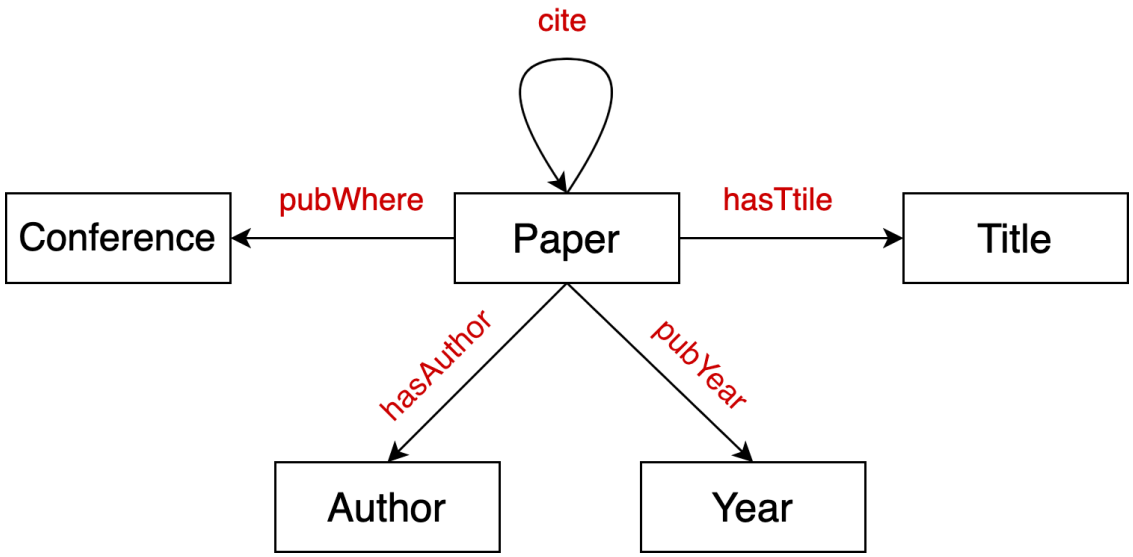
Knowledge in graph form:

- Capture entities, types, and relationships
- Nodes are **entities**
- Nodes are labeled with their **types**
- Edges between two nodes capture **relationships** between entities
- **KG is an example of a heterogeneous graph**



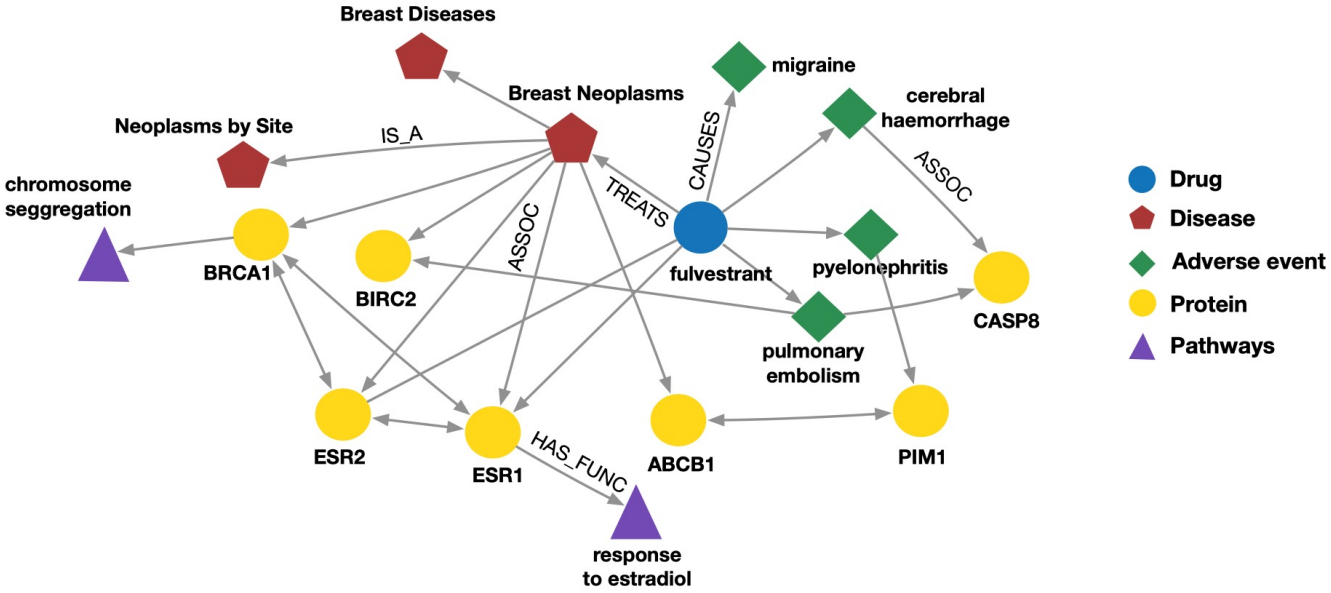
Example: Bibliographic Networks

- **Node types**: paper, title, author, conference, year
- **Relation types**: pubWhere, pubYear, hasTitle, hasAuthor, cite



Example: Bio Knowledge Graphs

- **Node types**: drug, disease, adverse event, protein, pathways
- **Relation types**: has_func, causes, assoc, treats, is_a



Knowledge Graphs in Practice

Examples of knowledge graphs

- Google Knowledge Graph
- Amazon Product Graph
- Meta Graph API
- IBM Watson
- Microsoft Satori
- Project Hanover/Literome
- LinkedIn Knowledge Graph
- Yandex Object Answer

Applications of Knowledge Graphs

- **Serving information:**

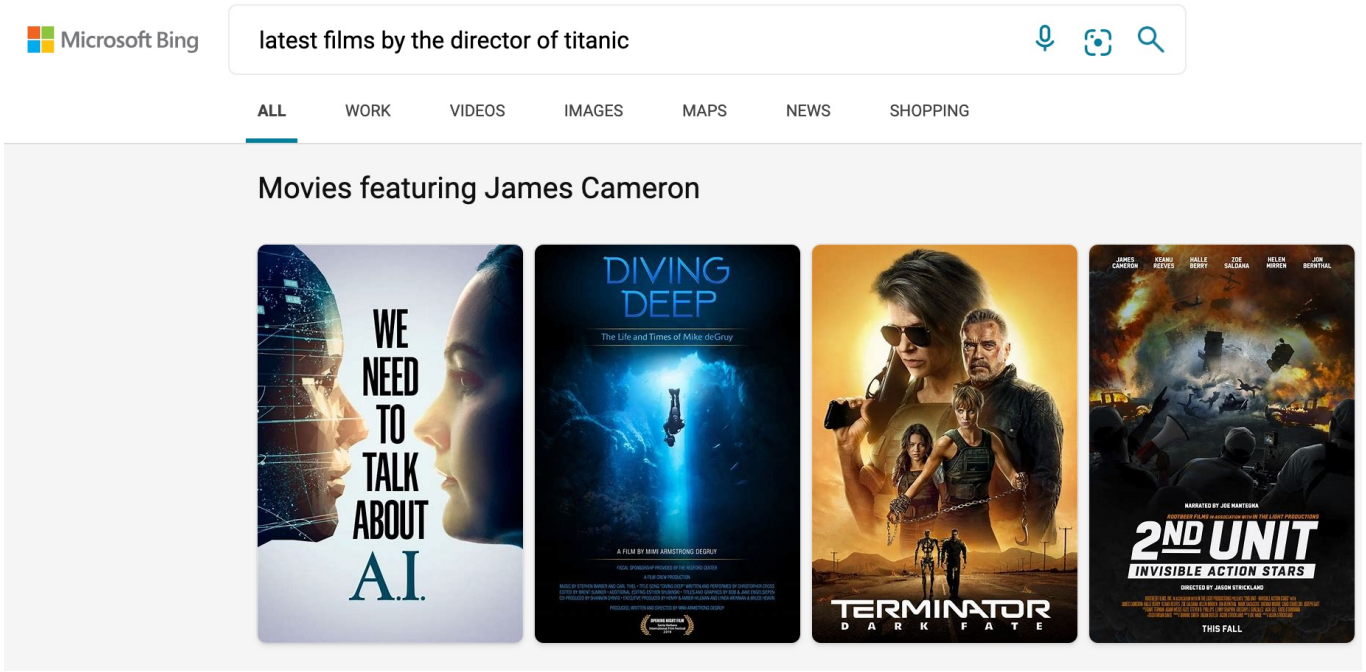
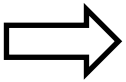


Image credit: Bing

Knowledge Graph Datasets

- **Publicly available KGs:**
  - FreeBase, Wikidata, Dbpedia, YAGO, NELL, etc.
- **Common characteristics:**
  - **Massive**: Millions of nodes and edges
  - **Incomplete**: Many true edges are missing

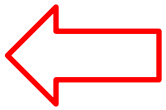
Given a massive KG, enumerating all the possible facts is intractable!



Can we predict plausible BUT missing links?

Example: Freebase

- **Freebase**
  - ~80 million **entities**
  - ~38K **relation types**
  - ~3 billion **facts/triples**



93.8% of persons from Freebase have no place of birth and 78.5% have no nationality!

- **Datasets:** FB15k/FB15k-237
  - A **complete** subset of Freebase, used by researchers to learn KG models

Dataset	Entities	Relations	Total Edges
FB15k	14,951	1,345	592,213
FB15k-237	14,505	237	310,079

[1] Paulheim, Heiko. "Knowledge graph refinement: A survey of approaches and evaluation methods." *Semantic web* 8.3 (2017): 489-508.
[2] Min, Bonan, et al. "Distant supervision for relation extraction with an incomplete knowledge base." *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. 2013.

Stanford CS224W: Knowledge Graph Completion

CS224W: Machine Learning with Graphs
Jure Leskovec, Stanford University
http://cs224w.stanford.edu

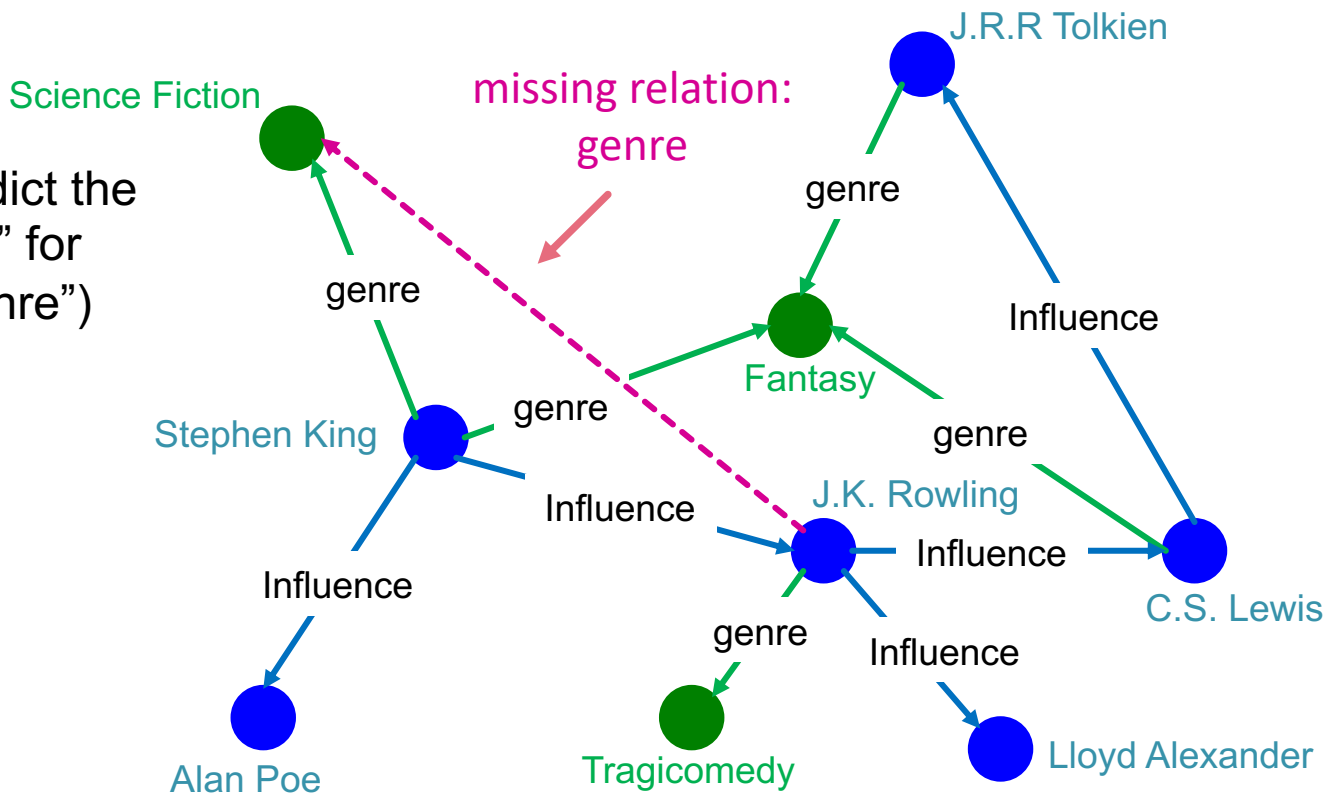


KG Completion Task

Given an enormous KG, can we complete the KG?

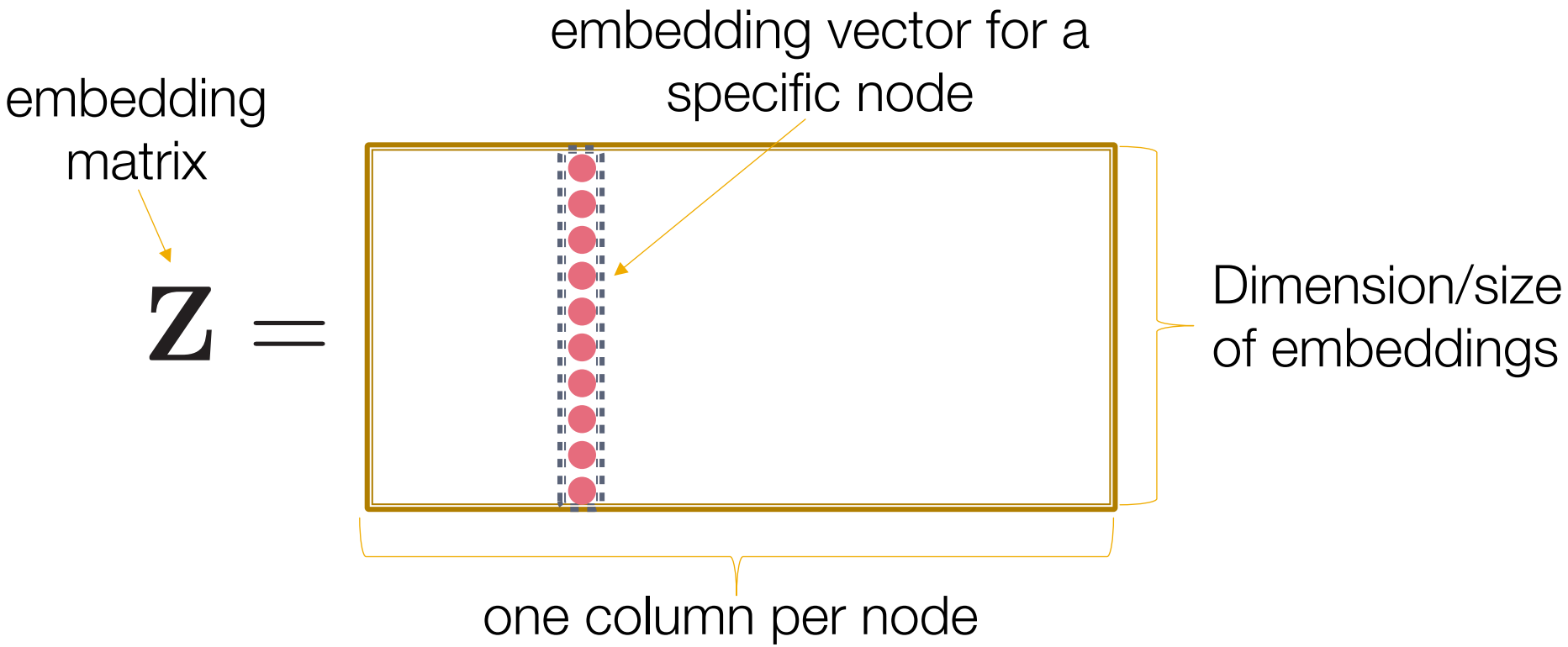
- For a given (**head**, **relation**), we predict missing **tails**.
  - (Note this is slightly different from link prediction task)

Example task: predict the tail "Science Fiction" for ("J.K. Rowling", "genre")



Recap: "Shallow" Encoding

- Simplest encoding approach: **encoder is just an embedding-lookup**

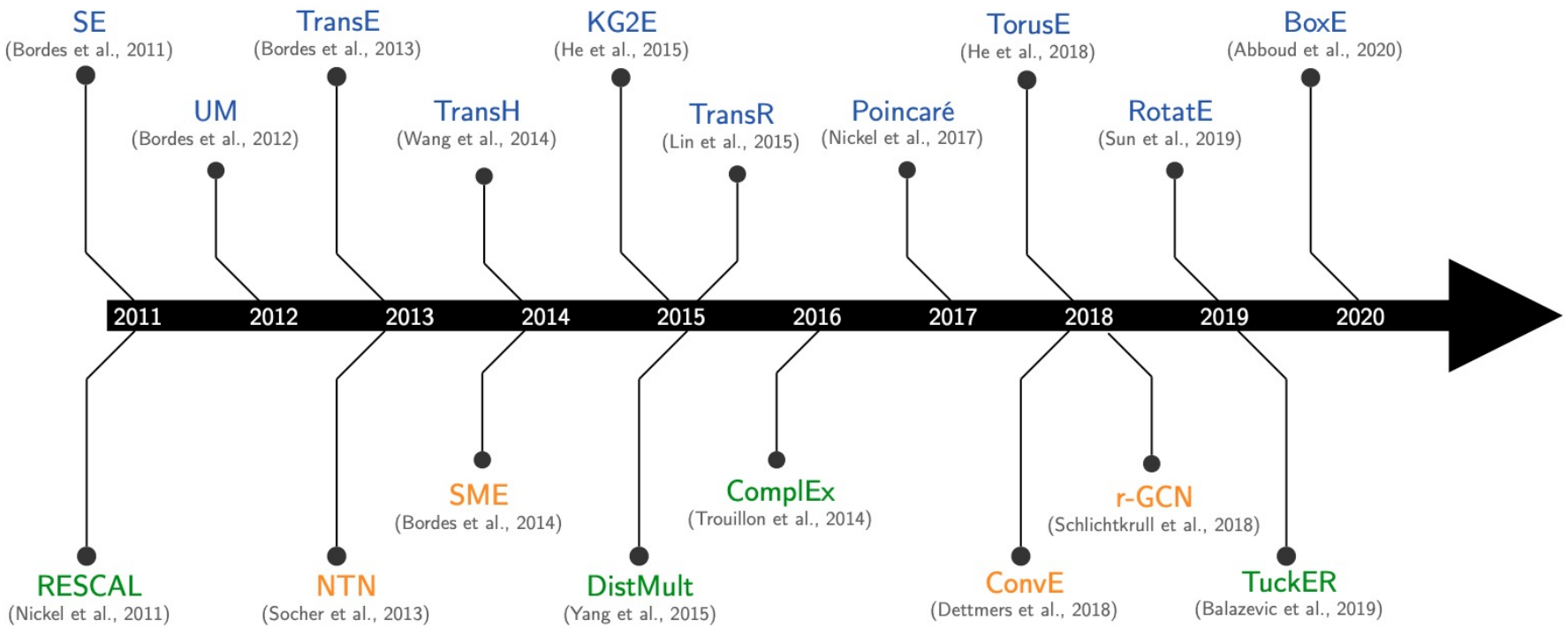


KG Representation

- Edges in KG are represented as **triples** (h, r, t)
  - head (h) has relation (r) with tail (t)
- **Key Idea:**
  - Model entities and relations in embedding space $\mathbb{R}^k$
    - Associate entities and relations with **shallow embeddings**
      - **Note we do not learn a GNN here!**
  - Given a triple (h, r, t), the goal is that the embedding of (h, r) **should be close** to the embedding of t.
    - How to embed (h, r)?
    - How to define score $f_r(h, t)$?
      - Score f_r is high if (h, r, t) exists, else f_r is low

Many KG Embedding Models

- Many KG embedding Models:



Today: Different Models

We are going to learn about different KG embedding models (shallow/transductive embs):

- Different models are…
  - …based on different geometric intuitions
  - …capture different types of relations (have different expressivity)

Model	Score	Embedding	Sym.	Antisym.	Inv.	Compos.	1-to-N
TransE	$-\|\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✗	✓	✓	✓	✗
TransR	$-\|\|\boldsymbol{M}_r\mathbf{h}+\mathbf{r}-\boldsymbol{M}_r\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t}\in\mathbb{R}^d$, $\mathbf{r}\in\mathbb{R}^k$, $\boldsymbol{M}_r\in\mathbb{R}^{k\times d}$	✓	✓	✓	✓	✓
DistMult	$<\mathbf{h},\mathbf{r},\mathbf{t}>$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✓	✗	✗	✗	✓
ComplEx	$\mathrm{Re}(<\mathbf{h},\mathbf{r},\bar{\mathbf{t}}>)$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{C}^k$	✓	✓	✓	✗	✓

Stanford CS224W: Knowledge Graph Completion: TransE

CS224W: Machine Learning with Graphs
Jure Leskovec, Stanford University
http://cs224w.stanford.edu

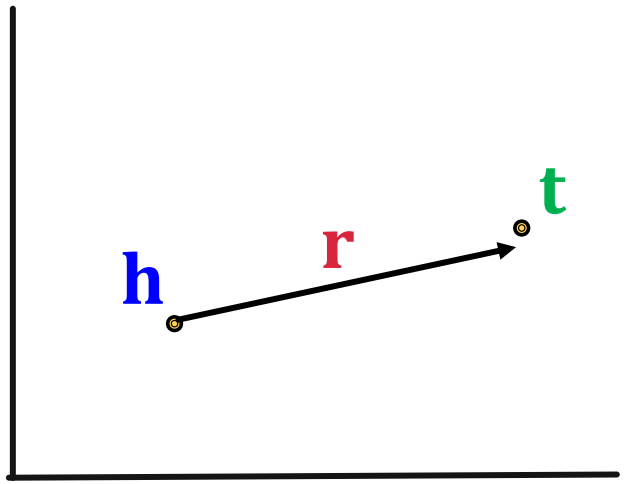


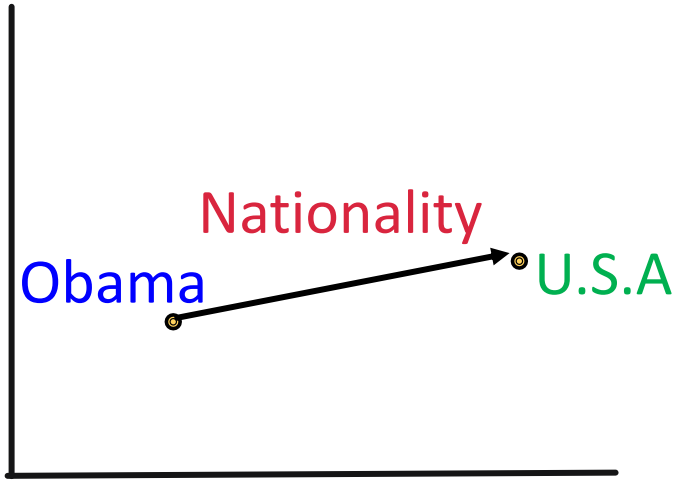
TransE

- **Intuition: Translation**
  For a triple (h, r, t), let $\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^k$ be embedding vectors.

  embedding vectors will appear in boldface
- **TransE:** $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$ if the given link exists else $\mathbf{h} + \mathbf{r} \neq \mathbf{t}$

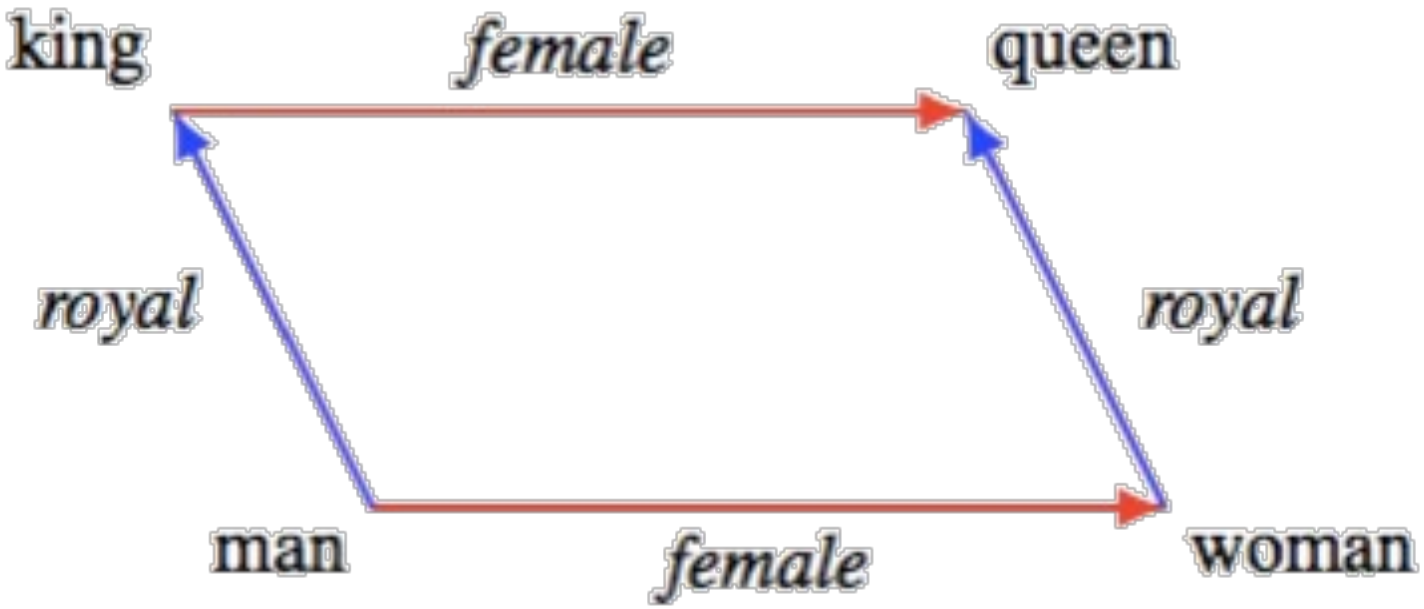
Entity scoring function: $f_r(h, t) = -||\mathbf{h} + \mathbf{r} - \mathbf{t}||$





TransE: Idea

- Entity embeddings



TransE: How to Learn

Algorithm 1 Learning TransE

input Training set $S = \{(h, r, t)\}$, entities and rel. sets E and R, margin γ, embeddings dim. k.

1: **initialize** $r \leftarrow \text{uniform}(-\frac{6}{\sqrt{k}}, \frac{6}{\sqrt{k}})$ for each $r \in R$
2: $r \leftarrow r / \|r\|$ for each $r \in R$
3: $\mathbf{e} \leftarrow \text{uniform}(-\frac{6}{\sqrt{k}}, \frac{6}{\sqrt{k}})$ for each entity $e \in E$
4: **loop**
5: $\mathbf{e} \leftarrow \mathbf{e} / \|\mathbf{e}\|$ for each entity $e \in E$
6: $S_{batch} \leftarrow$ sample(S, b) // sample a minibatch of size b
7: $T_{batch} \leftarrow \emptyset$ // initialize the set of pairs of triplets
8: **for** $(h, r, t) \in S_{batch}$ **do**
9: $(h', r, t') \leftarrow$ sample$(S'_{(h,r,t)})$ // sample a corrupted triplet
10: $T_{batch} \leftarrow T_{batch} \cup \{((h, r, t), (h', r, t'))\}$
11: **end for**
12: Update embeddings w.r.t.

$$\sum_{((h,r,t),(h',r,t')) \in T_{batch}} [\gamma + d(\boldsymbol{h} + r, \boldsymbol{t}) - d(\boldsymbol{h'} + r, \boldsymbol{t'})]_+$$

positive sample / negative sample

13: **end loop**

Initialize relations r and entities e uniformly, then normalize.
γ is the margin.

Sample triplet (h', r, t) that does not appear in the KG.

d represents distance (negative of score)

Contrastive loss: Favors lower distance (or higher score) for valid triplets, high distance (or lower score) for corrupted ones

Connectivity Patterns in KG

- **Relations in a heterogeneous KG have different properties:**
  - Example:
    - **Symmetry:** If the edge $(h, \text{"Roommate"}, t)$ exists in KG, then the edge $(t, \text{"Roommate"}, h)$ should also exist.
    - **Inverse relation:** If the edge $(h, \text{"Advisor"}, t)$ exists in KG, then the edge $(t, \text{"Advisee"}, h)$ should also exist.
- Can we **categorize** these relation patterns?
- **Are KG embedding methods (e.g., TransE) expressive enough to model these patterns?**

Four Relation Patterns

- **Symmetric (Antisymmetric) Relations:**

$$r(h,t) \Rightarrow r(t,h) \quad (r(h,t) \Rightarrow \neg r(t,h)) \quad \forall h,t$$

  - **Example**:
    - Symmetric: Family, Roommate
    - Antisymmetric: Hypernym (a word with a broader meaning: poodle vs. dog)
- **Inverse Relations:**

$$r_2(h,t) \Rightarrow r_1(t,h)$$

  - **Example** : (Advisor, Advisee)
- **Composition (Transitive) Relations:**

$$r_1(x,y) \wedge r_2(y,z) \Rightarrow r_3(x,z) \quad \forall x,y,z$$

  - **Example**: My mother's husband is my father.
- **1-to-N relations:**

$$r(h,t_1), r(h,t_2), \ldots, r(h,t_n) \text{ are all True.}$$

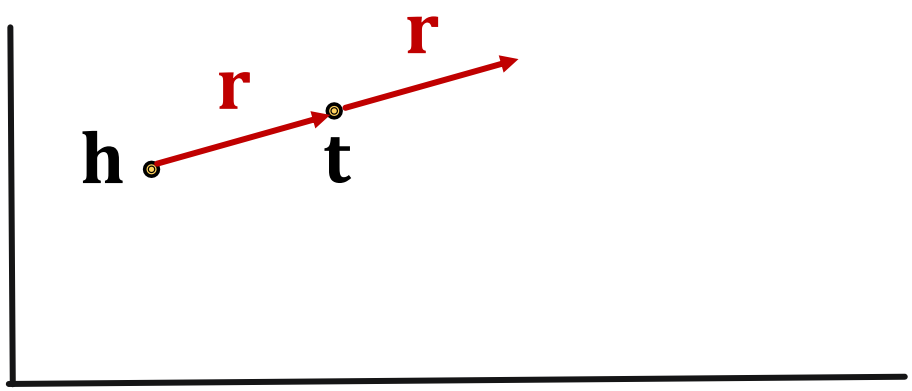
  - **Example**: r is "StudentsOf"

Antisymmetric Relations in TransE

- **Antisymmetric Relations:**

$$r(h,t) \Rightarrow \neg r(t,h) \quad \forall h,t$$

  - **Example**: Hypernym (a word with a broader meaning: poodle vs. dog)
- **TransE** can model antisymmetric relations ✓
  - $\mathbf{h} + \mathbf{r} = \mathbf{t}$, but $\mathbf{t} + \mathbf{r} \neq \mathbf{h}$

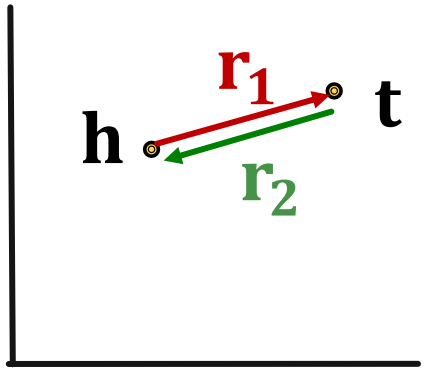


Inverse Relations in TransE

- **Inverse Relations:**

$$r_2(h,t) \Rightarrow r_1(t,h)$$

  - **Example** : (Advisor, Advisee)
- **TransE** can model inverse relations ✓
  - $\mathbf{h} + \mathbf{r}_2 = \mathbf{t}$, we can set $\mathbf{r}_1 = -\mathbf{r}_2$



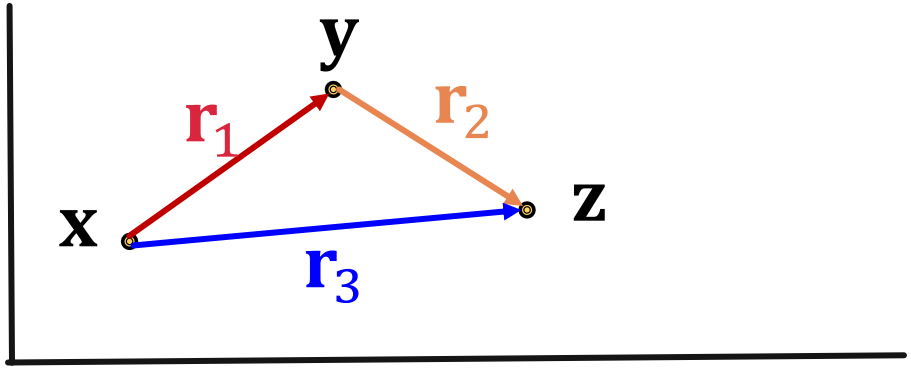
Composition in TransE

- **Composition (Transitive) Relations:**

$$r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z) \quad \forall x, y, z$$

  - **Example**: My mother's husband is my father.
- **TransE** can model composition relations ✓

$$\mathbf{r}_3 = \mathbf{r}_1 + \mathbf{r}_2$$



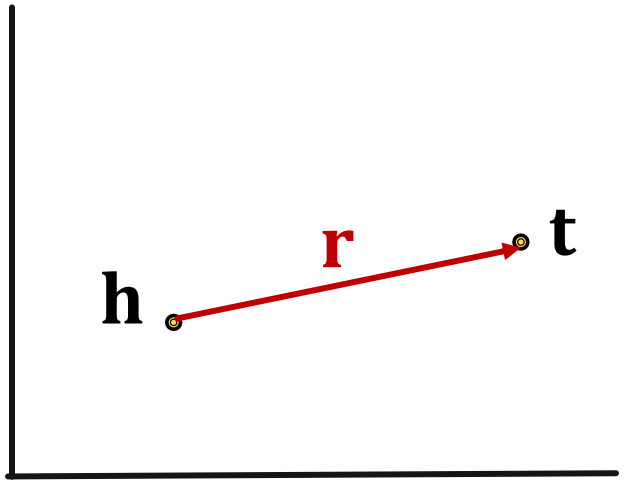
Limitation: Symmetric Relations

- **Symmetric Relations:**

$$r(h,t) \Rightarrow r(t,h) \quad \forall h,t$$

  - **Example**: Family, Roommate
- **TransE cannot** model symmetric relations ✘

only if $\mathbf{r} = 0,\ \mathbf{h} = \mathbf{t}$

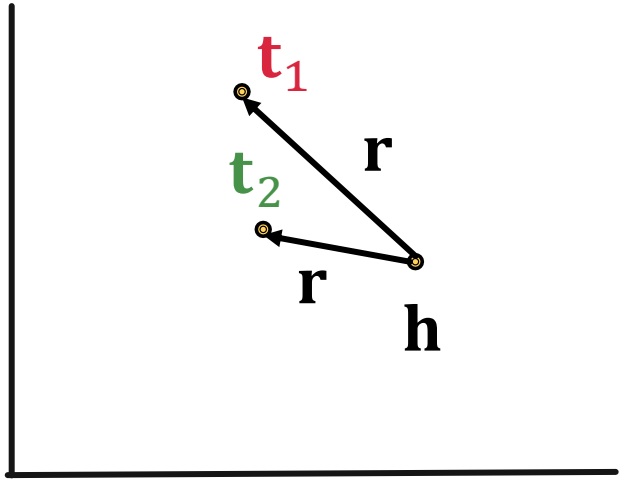


For all h,t that satisfy $r(h,t)$, $r(t,h)$ is also True, which means $\|\mathbf{h}+\mathbf{r}-\mathbf{t}\| = 0$ and $\|\mathbf{t}+\mathbf{r}-\mathbf{h}\| = 0$. Then $\mathbf{r} = 0$ and $\mathbf{h} = \mathbf{t}$, however h and t are two different entities and should be mapped to different locations.

Limitation: 1-to-N Relations

- **1-to-N Relations:**
  - **Example**: (h, r, t_1) and (h, r, t_2) both exist in the knowledge graph, e.g., r is "StudentsOf"
- **TransE cannot** model 1-to-N relations ✘
  - $\mathbf{t}_1$ and $\mathbf{t}_2$ will map to the same vector, although they are different entities

- $\mathbf{t}_1 = \mathbf{h} + \mathbf{r} = \mathbf{t}_2$
- $\mathbf{t}_1 \neq \mathbf{t}_2$ contradictory!



Today: KG Completion Models

What we learned so far:

Model	Score	Embedding	Sym.	Antisym.	Inv.	Compos.	1-to-N
TransE	$-\|\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✗	✓	✓	✓	✗

Stanford CS224W: Knowledge Graph Completion: TransR

CS224W: Machine Learning with Graphs
Jure Leskovec, Stanford University
http://cs224w.stanford.edu



TransR

- **TransE models translation of any relation in the same embedding space.**

- **Can we design a new space for each relation and do translation in relation-specific space?**

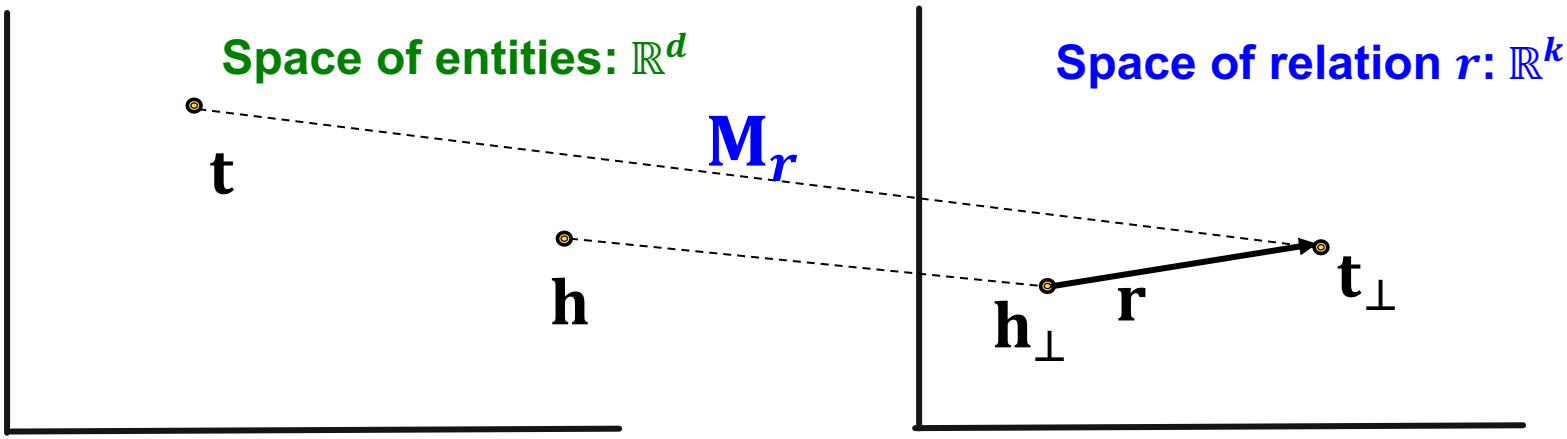
- **TransR**: model **entities** as vectors in the entity space $\mathbb{R}^d$ and model each **relation** as vector in relation space $\mathbf{r} \in \mathbb{R}^k$ with $\mathbf{M}_r \in \mathbb{R}^{k \times d}$ as the projection matrix.

TransR

- **TransR**: model **entities** as vectors in the entity space $\mathbb{R}^d$ and model each **relation** as vector in relation space $\mathbf{r} \in \mathbb{R}^k$ with $\mathbf{M}_r \in \mathbb{R}^{k \times d}$ as the **projection matrix**.

Use $\mathbf{M}_r$ to project from entity space $\mathbb{R}^d$ to relation space $\mathbb{R}^k$!

- $\mathbf{h}_\perp = \mathbf{M}_r\mathbf{h},\ \mathbf{t}_\perp = \mathbf{M}_r\mathbf{t}$
- **Score function**: $f_r(h,t) = -||\mathbf{h}_\perp + \mathbf{r} - \mathbf{t}_\perp||$



Symmetric Relations in TransR

- **Symmetric Relations:**

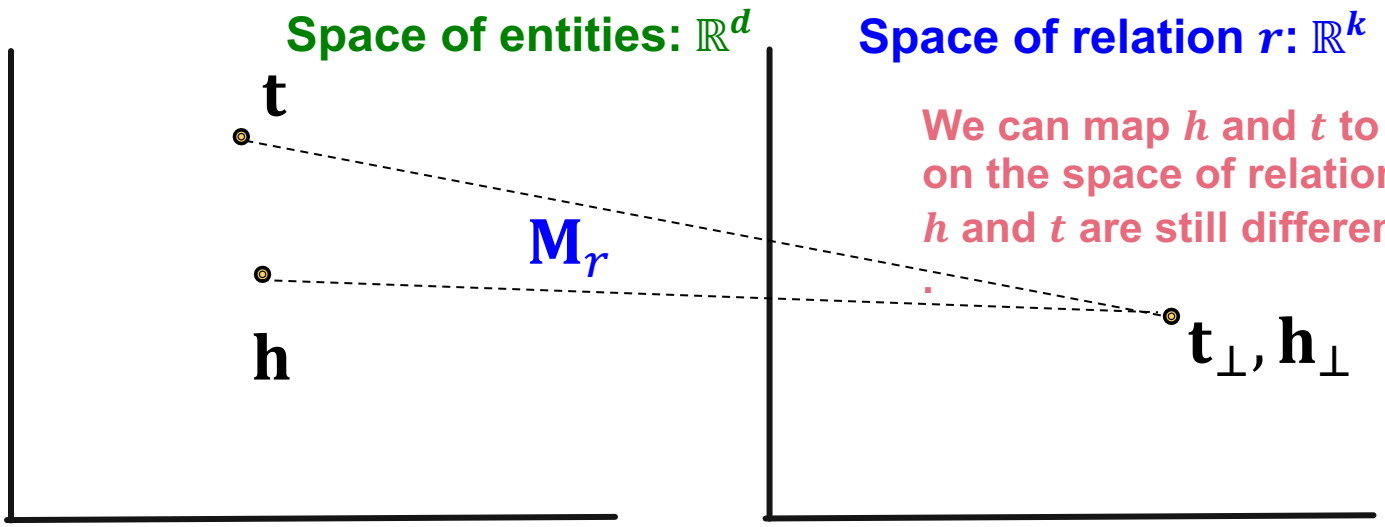
$$r(h,t) \Rightarrow r(t,h) \quad \forall h,t$$

  - **Example**: Family, Roommate

Note different symmetric relations may have different $\mathbf{M}_r$

- **TransR** can model symmetric relations

$$\mathbf{r} = 0,\ \mathbf{h}_\perp = \mathbf{M}_r\mathbf{h} = \mathbf{M}_r\mathbf{t} = \mathbf{t}_\perp \checkmark$$



We can map h and t to the same location on the space of relations r. Then $r = 0$.
h and t are still different in the entity space,

Antisymmetric Relations in TransR

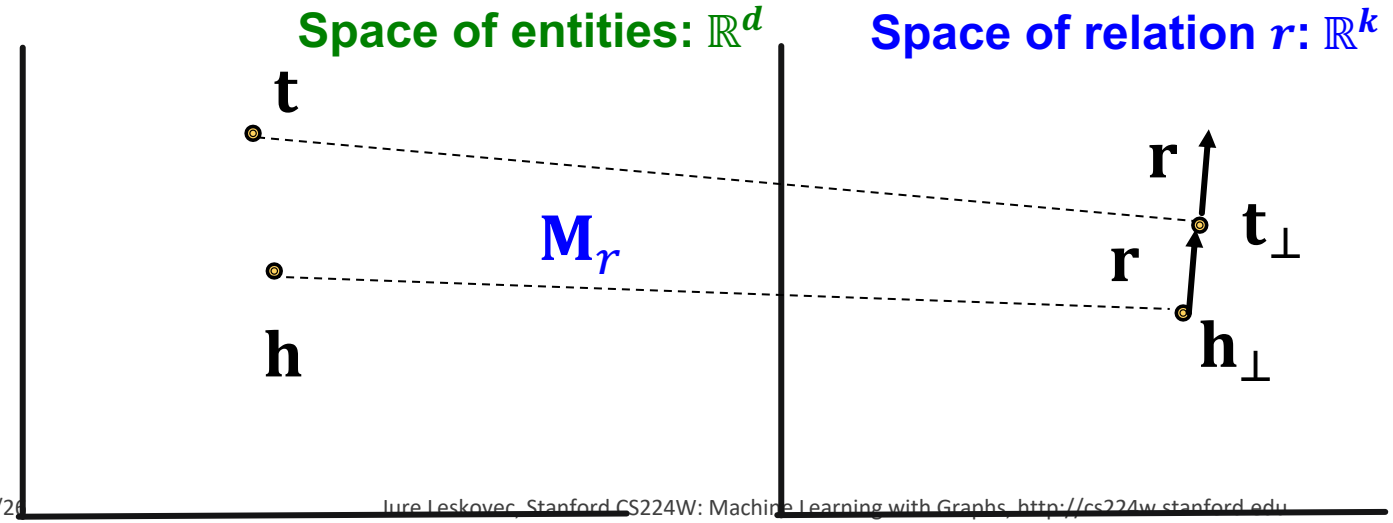
- **Antisymmetric Relations:**

$$r(h,t) \Rightarrow \neg r(t,h) \quad \forall h,t$$

  - **Example**: Hypernym
- **TransR** can model antisymmetric relations:

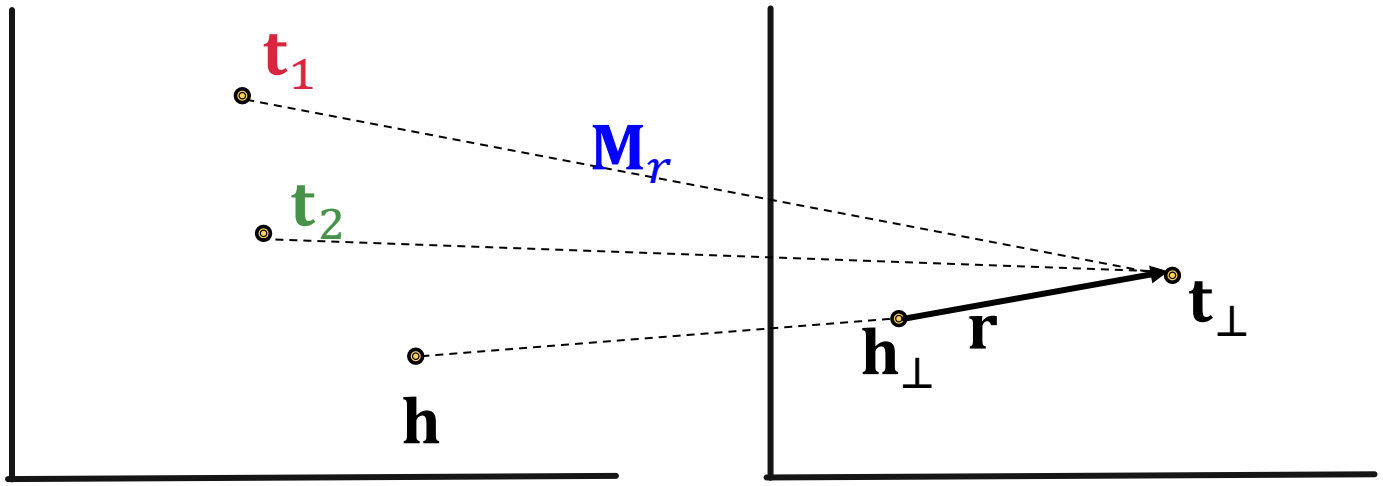
$$\mathbf{r} \neq 0, \mathbf{M}_r\mathbf{h} + \mathbf{r} = \mathbf{M}_r\mathbf{t},$$

$$\text{Then } \mathbf{M}_r\mathbf{t} + \mathbf{r} \neq \mathbf{M}_r\mathbf{h} \checkmark$$



1-to-N Relations in TransR

- **1-to-N Relations:**
  - **Example**: If (h, r, t_1) and (h, r, t_2) exist in the knowledge graph.
- **TransR** can model 1-to-N relations ✓
  - We can learn $\mathbf{M}_r$ so that $\mathbf{t}_\perp = \mathbf{M}_r \mathbf{t}_1 = \mathbf{M}_r \mathbf{t}_2$
  - Note that $\mathbf{t}_1$ does not need to be equal to $\mathbf{t}_2$!



Inverse Relations in TransR

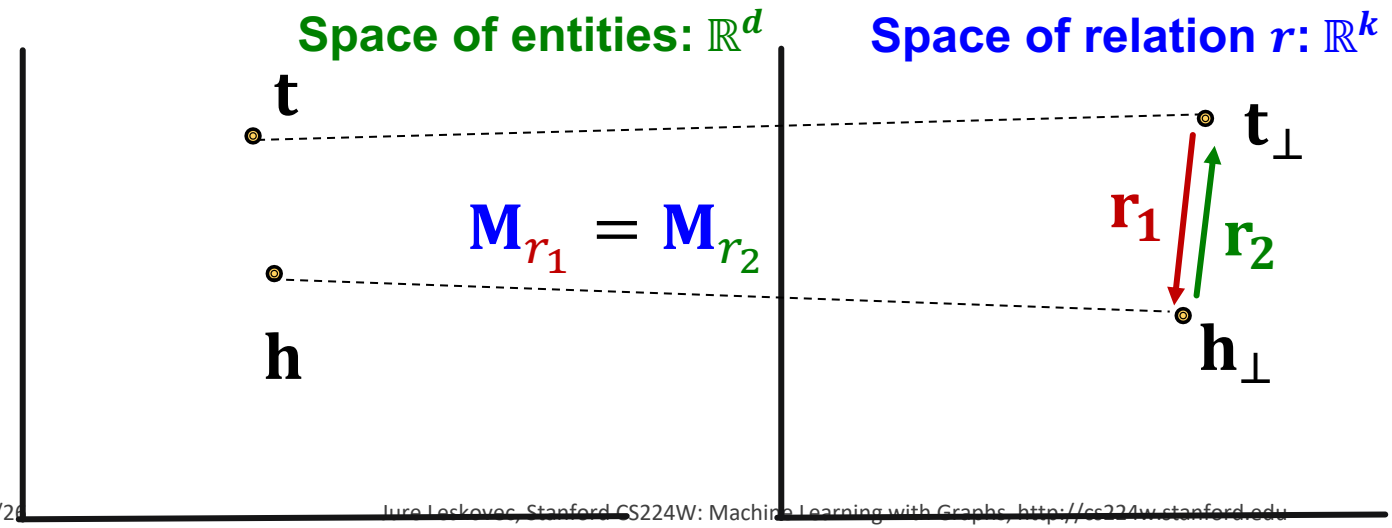
- **Inverse Relations:**

$$r_2(h,t) \Rightarrow r_1(t,h)$$

  - **Example** : (Advisor, Advisee)
- **TransR** can model inverse relations

$$\mathbf{r}_2 = -\mathbf{r}_1, \mathbf{M}_{r_1} = \mathbf{M}_{r_2}$$

Then $\mathbf{M}_{r_1}\mathbf{t} + \mathbf{r}_1 = \mathbf{M}_{r_1}\mathbf{h}$ and $\mathbf{M}_{r_2}\mathbf{h} + \mathbf{r}_2 = \mathbf{M}_{r_2}\mathbf{t}$ ✓



Composition Relations in TransR

- **Composition Relations:**

$$r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z) \quad \forall x, y, z$$

  - **Example**: My mother's husband is my father.
- **TransR** can model composition relations

High-level intuition: TransR models a triple with linear functions. Linear functions are chainable!

- If f(x) and g(x) are linear, then f(g(x)) is also linear:
  - Let: f(x)=a·x+b, g(x)=c·x+d: then f(g(x))= a(c·x+d)+b.

Composition Relations in TransR

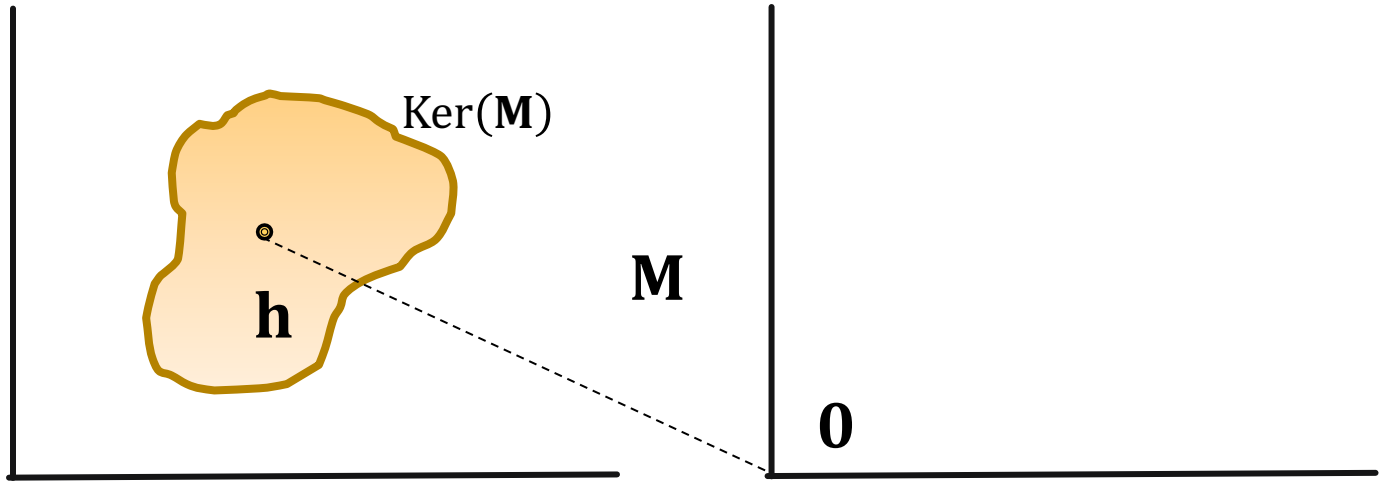
- **Composition Relations:**

$$r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z) \quad \forall x, y, z$$

Background:

Def: Kernel space of a matrix $\mathbf{M}$:

$$\mathbf{h} \in \mathrm{Ker}(\mathbf{M}), \text{ then } \mathbf{M} \cdot \mathbf{h} = \mathbf{0}$$



Composition Relations in TransR

- **Composition Relations:**

$$r_1(x,y) \wedge r_2(y,z) \Rightarrow r_3(x,z) \quad \forall x,y,z$$

Assume $\mathbf{M}_{r_1}\mathbf{g}_1 = \mathbf{r}_1$ and $\mathbf{M}_{r_2}\mathbf{g}_2 = \mathbf{r}_2$

- For $r_1(x,y)$:

$r_1(x,y)$ exists $\Rightarrow \mathbf{M}_{r_1}\mathbf{x} + \mathbf{r_1} = \mathbf{M}_{r_1}\mathbf{y} \Rightarrow \mathbf{M}_{r_1}(\boldsymbol{y}-\boldsymbol{x}) = \boldsymbol{r}_1$

$$\mathbf{y} - \mathbf{x} \in \mathbf{g}_1 + \mathrm{Ker}(\mathbf{M}_{r_1}) \Rightarrow \mathbf{y} \in \mathbf{x} + \mathbf{g_1} + \mathrm{Ker}(\mathbf{M}_{r_1})$$

- Same for $r_2(y,z)$:

$r_2(y,z)$ exists $\Rightarrow \mathbf{M}_{r_2}\mathbf{y} + \mathbf{r_2} = \mathbf{M}_{r_2}\mathbf{z} \Rightarrow$

$$\mathbf{z} - \mathbf{y} \in \mathbf{g}_2 + \mathrm{Ker}(\mathbf{M}_{r_2}) \Rightarrow \mathbf{z} \in \mathbf{y} + \mathbf{g_2} + \mathrm{Ker}(\mathbf{M}_{r_2})$$

- Then, we have

$$\mathbf{z} \in \mathbf{x} + \mathbf{g_1} + \mathbf{g_2} + \mathrm{Ker}(\mathbf{M}_{r_1}) + \mathrm{Ker}(\mathbf{M}_{r_2})$$

Composition Relations in TransR

- **Composition Relations:**

$$r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z) \quad \forall x, y, z$$

We have $\mathbf{z} \in \mathbf{x} + \mathbf{g_1} + \mathbf{g_2} + \mathrm{Ker}(\mathbf{M}_{r_1}) + \mathrm{Ker}(\mathbf{M}_{r_2})$

- Construct $\mathbf{M}_{r_3}$, s.t.
  $\mathrm{Ker}(\mathbf{M}_{r_3}) = \mathrm{Ker}(\mathbf{M}_{r_1}) + \mathrm{Ker}(\mathbf{M}_{r_2})$

- **Since:**
  - $\dim\left(\mathrm{Ker}(\mathbf{M}_{r_3})\right) \geq \dim\left(\mathrm{Ker}(\mathbf{M}_{r_1})\right)$
  - $\mathbf{M}_{r_3}$ has the same shape as $\mathbf{M}_{r1}$

  we know $\mathbf{M}_{r_3}$ exists!

- Set $\mathbf{r}_3 = \mathbf{M}_{r_3}(\mathbf{g}_1 + \mathbf{g}_2)$

- We have $\mathbf{M}_{r_3}\mathbf{x} + \mathbf{r_3} = \mathbf{M}_{r_3}\mathbf{z}$

Today: KG Completion Models

What we learned so far:

Model	Score	Embedding	Sym.	Antisym.	Inv.	Compos.	1-to-N
TransE	$-\|\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✗	✓	✓	✓	✗
TransR	$-\|\|\boldsymbol{M}_r\mathbf{h}+\mathbf{r}-\boldsymbol{M}_r\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t}\in\mathbb{R}^d$, $\mathbf{r}\in\mathbb{R}^k$, $\boldsymbol{M}_r\in\mathbb{R}^{k\times d}$	✓	✓	✓	✓	✓

Stanford CS224W: Knowledge Graph Completion: DistMult

CS224W: Machine Learning with Graphs
Jure Leskovec, Stanford University
http://cs224w.stanford.edu



New Idea: Bilinear Modeling

- **So far:** The scoring function $f_r(h, t)$ is **negative of L1 / L2 distance** in **TransE** and **TransR**
- **Idea: Use bilinear** modeling:

  Score function: $f_r(h, t) = h \cdot A \cdot t$

  $\mathbf{h}, \mathbf{t} \in \mathbb{R}^k, \mathbf{A} \in \mathbb{R}^{k \times k}$

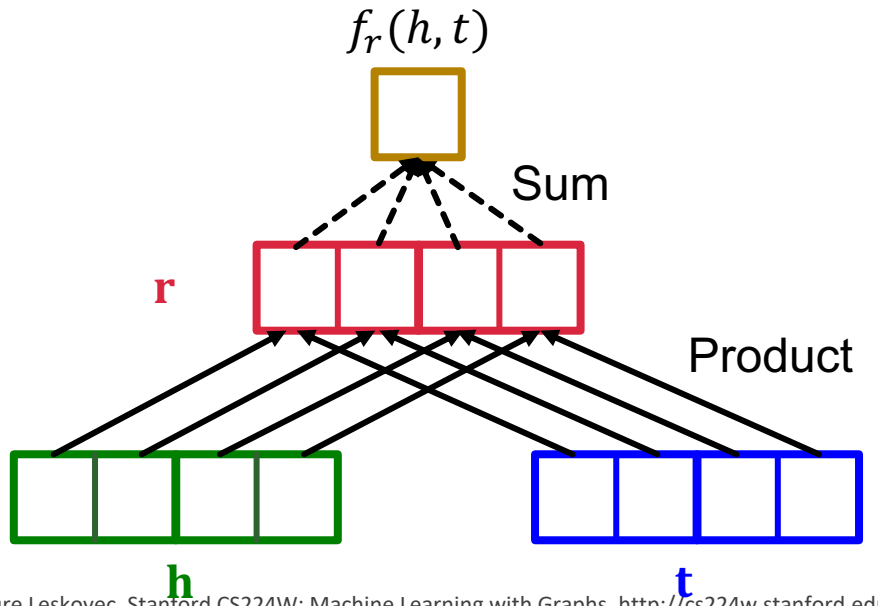
- **Problem: Too general and prone to overfitting**
  - Matrix A is too expressive
- **Fix: Limit A to be diagonal**
  - **This is called DistMult**

New Idea: Bilinear Modeling

- **DistMult**: Entities & relations are vectors in $\mathbb{R}^k$
- **Score function**:

$$f_r(h,t) =< \mathbf{h}, \mathbf{r}, \mathbf{t} >= \sum_i \mathbf{h}_i \cdot \mathbf{r}_i \cdot \mathbf{t}_i$$

- $\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^k$



DistMult

- **DistMult**: Entities and relations using vectors in $\mathbb{R}^k$
- **Score function:** $f_r(h,t) = < \mathbf{h}, \mathbf{r}, \mathbf{t} > = \sum_i \mathbf{h}_i \cdot \mathbf{r}_i \cdot \mathbf{t}_i$
  - $\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^k$
- **Intuition of the score function:** Can be viewed as a cosine similarity between $\mathbf{h} \cdot \mathbf{r}$ and $\mathbf{t}$

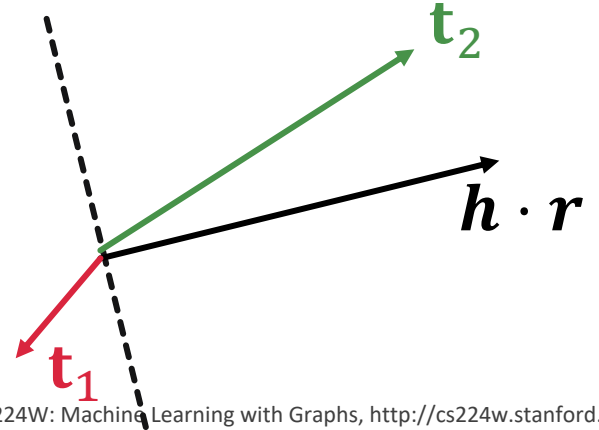
  where $\mathbf{h} \cdot \mathbf{r}$ is defined as $[\boldsymbol{h} \cdot \boldsymbol{r}]_i = \boldsymbol{h}_i \cdot \boldsymbol{r}_i$
- **Example:**

Hadamard product

$$f_r(h, t_1) < 0, \qquad f_r(h, t_2) > 0$$

$$\cos(\mathbf{h} \cdot \mathbf{r}, \mathbf{t}) = \frac{< \mathbf{h}, \mathbf{r}, \mathbf{t} >}{\|\mathbf{h} \cdot \mathbf{r}\| \|\mathbf{t}\|}$$

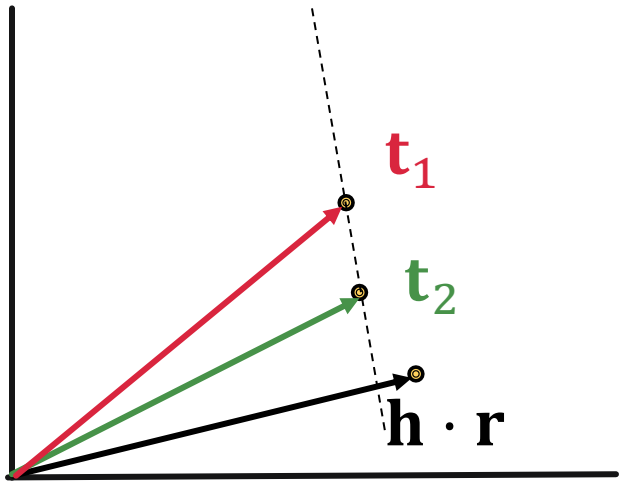
$$< \mathbf{h}, \mathbf{r}, \mathbf{t} > = \cos(\mathbf{h} \cdot \mathbf{r}, \mathbf{t}) \|\mathbf{h} \cdot \mathbf{r}\| \|\mathbf{t}\|$$



1-to-N Relations in DistMult

- **1-to-N Relations:**
  - **Example**: If (h, r, t_1) and (h, r, t_2) exist in the knowledge graph
- **DistMult** can model 1-to-N relations ✓

$$< \mathbf{h}, \mathbf{r}, \mathbf{t}_1 > = < \mathbf{h}, \mathbf{r}, \mathbf{t}_2 >$$



Symmetric Relations in DistMult

- **Symmetric Relations:**

$$r(h,t) \Rightarrow r(t,h) \quad \forall h,t$$

  - **Example**: Family, Roommate
- **DistMult** can naturally model symmetric relations ✓

$$f_r(h,t) = < \mathbf{h}, \mathbf{r}, \mathbf{t} > = \sum_i \mathbf{h}_i \cdot \mathbf{r}_i \cdot \mathbf{t}_i =$$

$$< \mathbf{t}, \mathbf{r}, \mathbf{h} > = f_r(t,h)$$

Due to the commutative property of multiplication.

Limitation: Antisymmetric Relations

- **Antisymmetric Relations:**

$$r(h,t) \Rightarrow \neg r(t,h) \quad \forall h,t$$

  - **Example**: Hypernym

- **DistMult cannot** model antisymmetric relations

$$f_r(h,t) = < \mathbf{h}, \mathbf{r}, \mathbf{t} > = < \mathbf{t}, \mathbf{r}, \mathbf{h} > = f_r(t,h)$$ ✘

  - $r(h,t)$ and $r(t,h)$ always have same score!

DistMult cannot differentiate between head entity and tail entity! This means that all relations are modelled as symmetric regardless, i.e., even anti-symmetric relations will be represented as symmetric.

Limitation: Inverse Relations

- **Inverse Relations:**

$$r_2(h,t) \Rightarrow r_1(t,h)$$

  - **Example** : (Advisor, Advisee)
- **DistMult cannot** model inverse relations ✘
  - Assume DistMult does model inverse relations:

$$f_{r_2}(h,t) =< \mathbf{h}, \mathbf{r_2}, \mathbf{t} > =< \mathbf{t}, \mathbf{r_1}, \mathbf{h} >= f_{r_1}(t,h)$$

  - For example, $\mathbf{r_2} = \mathbf{r_1}$ solves this (there are also exist solutions $\mathbf{r_2} \neq \mathbf{r_1}$)
  - But semantically this does not make sense: **The embedding of "Advisor" relation should not be the same as "Advisee" relation.**

Limitation: Composition Relations

- **Composition Relations:**

$$r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z) \quad \forall x, y, z$$

  - **Example**: My mother's husband is my father.

- **DistMult cannot** model composition of relations ✘
  - **Intuition:** Because dot product is commutative (a·b= b·a) **DistMult** does not distinguish between head and tail entities, so it cannot model composition.

Today: KG Completion Models

What we learned so far:

Model	Score	Embedding	Sym.	Antisym.	Inv.	Compos.	1-to-N
TransE	$-\|\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✗	✓	✓	✓	✗
TransR	$-\|\|\boldsymbol{M}_r\mathbf{h}+\mathbf{r}-\boldsymbol{M}_r\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t}\in\mathbb{R}^d$, $\mathbf{r}\in\mathbb{R}^k$, $\boldsymbol{M}_r\in\mathbb{R}^{k\times d}$	✓	✓	✓	✓	✓
DistMult	$<\mathbf{h},\mathbf{r},\mathbf{t}>$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✓	✗	✗	✗	✓

Stanford CS224W: Knowledge Graph Completion: ComplEx

CS224W: Machine Learning with Graphs

Jure Leskovec, Stanford University

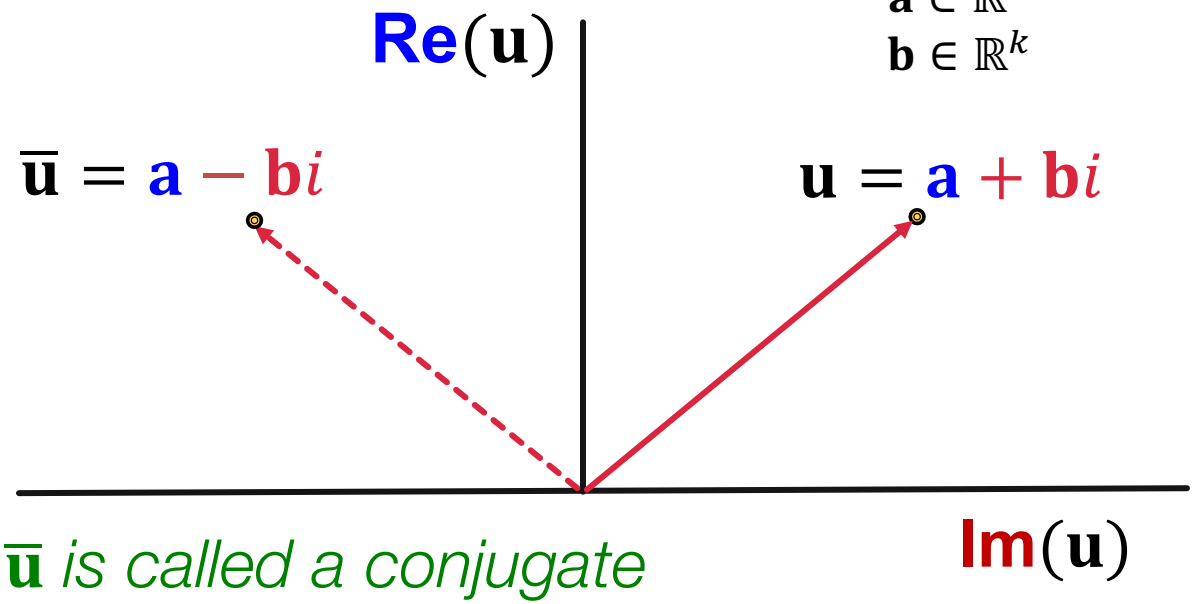
http://cs224w.stanford.edu



ComplEx

- Based on Distmult, **ComplEx** embeds entities and relations in **Complex vector space**
- **ComplEx**: model entities and relations using vectors in $\mathbb{C}^k$

$\mathbf{u} \in \mathbb{C}^k$
$\mathbf{a} \in \mathbb{R}^k$
$\mathbf{b} \in \mathbb{R}^k$

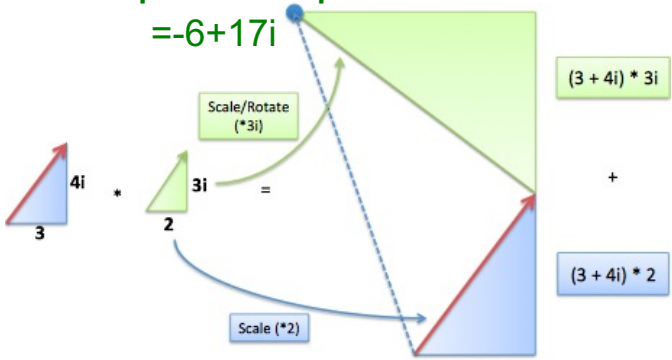


ū *is called a conjugate*

Complex multiplication:

(a + ib) (c + id) = (ac - bd) + i(ad + bc)

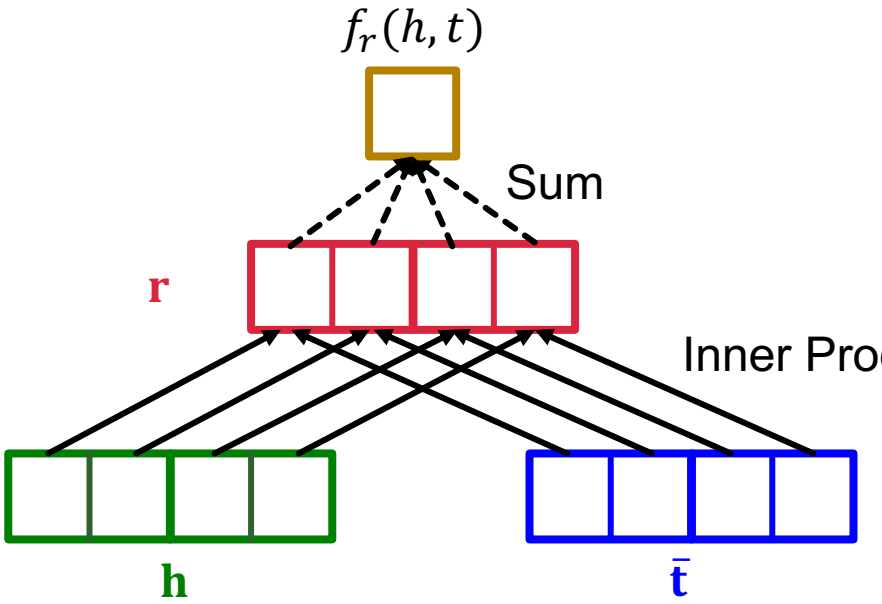
Example multiplication:



ComplEx

- Based on Distmult, **ComplEx** embeds entities and relations in **Complex vector space**
- **ComplEx**: model entities and relations using vectors in $\mathbb{C}^k$
- **Score function** $f_r(h,t) = \mathrm{Re}(\sum_i \mathbf{h}_i \cdot \mathbf{r}_i \cdot \bar{\mathbf{t}}_i)$

$$= \langle \mathrm{Re}(\mathbf{h}_i), \mathrm{Re}(\mathbf{r}_i), \mathrm{Re}(\mathbf{t}_i) \rangle + \langle \mathrm{Re}(\mathbf{h}_i), \mathrm{Im}(\mathbf{r}_i), \mathrm{Im}(\mathbf{t}_i) \rangle + \langle \mathrm{Im}(\mathbf{h}_i), \mathrm{Re}(\mathbf{r}_i), \mathrm{Im}(\mathbf{t}_i) \rangle - \langle \mathrm{Im}(\mathbf{h}_i), \mathrm{Im}(\mathbf{r}_i), \mathrm{Re}(\mathbf{t}_i) \rangle$$



ComplEx Score Function

$$f_r(h,t) = \text{Re}\left(\sum_i \mathbf{h}_i \cdot \mathbf{r}_i \cdot \bar{\mathbf{t}}_i\right)$$

$$= \sum_i \text{Re}(\mathbf{h}_i \cdot \mathbf{r}_i \cdot \bar{\mathbf{t}}_i)$$

$$= \sum_i \text{Re}((\text{Re}(\mathbf{h}_i) + i\text{Im}(\mathbf{h}_i)) \cdot (\text{Re}(\mathbf{r}_i) + i\text{Im}(\mathbf{r}_i)) \cdot (\text{Re}(\mathbf{t}_i) - i\text{Im}(\mathbf{t}_i)))$$

$$= \sum_i \begin{array}{l} \text{Re}(\mathbf{h}_i)\text{Re}(\mathbf{r}_i)\text{Re}(\mathbf{t}_i) + \text{Re}(\mathbf{h}_i)\text{Im}(\mathbf{r}_i)\text{Im}(\mathbf{t}_i) \\ +\text{Im}(\mathbf{h}_i)\text{Re}(\mathbf{r}_i)\text{Im}(\mathbf{t}_i) - \text{Im}(\mathbf{h}_i)\text{Im}(\mathbf{r}_i)\text{Re}(\mathbf{t}_i) \end{array}$$

$$= \langle \text{Re}(\mathbf{h}_i), \text{Re}(\mathbf{r}_i), \text{Re}(\mathbf{t}_i)\rangle + \langle \text{Re}(\mathbf{h}_i), \text{Im}(\mathbf{r}_i), \text{Im}(\mathbf{t}_i)\rangle$$
$$+ \langle \text{Im}(\mathbf{h}_i), \text{Re}(\mathbf{r}_i), \text{Im}(\mathbf{t}_i)\rangle - \langle \text{Im}(\mathbf{h}_i), \text{Im}(\mathbf{r}_i), \text{Re}(\mathbf{t}_i)\rangle$$

Antisymmetric Relations in ComplEx

- **Antisymmetric Relations:**

$$r(h,t) \Rightarrow \neg r(t,h) \quad \forall h,t$$

  - **Example**: Hypernym
- **ComplEx** can model antisymmetric relations ✓
  - The model is expressive enough to learn
    - **High** $f_r(h,t) = \text{Re}(\sum_i \mathbf{h}_i \cdot \mathbf{r}_i \cdot \bar{\mathbf{t}}_i)$
    - **Low** $f_r(t,h) = \text{Re}(\sum_i \boldsymbol{t}_i \cdot \mathbf{r}_i \cdot \overline{\boldsymbol{h}}_i)$

  Due to the asymmetric modeling using complex conjugate.

Symmetric Relations in ComplEx

- **Symmetric Relations:**

$$r(h,t) \Rightarrow r(t,h) \quad \forall h,t$$

  - **Example**: Family, Roommate
- **ComplEx** can model symmetric relations ✓
  - When $\mathrm{Im}(\mathbf{r}) = 0$, we have
    - $f_r(h,t) = \mathrm{Re}(\sum_i \mathbf{h}_i \cdot \mathbf{r}_i \cdot \bar{\mathbf{t}}_i) = \sum_i \mathrm{Re}(\mathbf{r}_i \cdot \mathbf{h}_i \cdot \bar{\mathbf{t}}_i)$ $= \sum_i \mathbf{r}_i \cdot \mathrm{Re}(\mathbf{h}_i \cdot \bar{\mathbf{t}}_i) = \sum_i \mathbf{r}_i \cdot \mathrm{Re}\left(\bar{\mathbf{h}}_i \cdot \mathbf{t}_i\right) = \sum_i \mathrm{Re}\left(\mathbf{r}_i \cdot \bar{\mathbf{h}}_i \cdot \mathbf{t}_i\right) = f_r(t,h)$

Inverse Relations in ComplEx

- Inverse Relations:

$$r_2(h,t) \Rightarrow r_1(t,h)$$

  - **Example** : (Advisor, Advisee)
- **ComplEx** can model inverse relations ✓
  - $\mathbf{r}_1 = \bar{\mathbf{r}}_2$
  - Complex conjugate of

$\mathbf{r}_2 = \underset{\mathbf{r}}{\text{argmax}}\, \text{Re}(< \mathbf{h}, \mathbf{r}, \bar{\mathbf{t}} >)$ is exactly

$\mathbf{r}_1 = \underset{\mathbf{r}}{\text{argmax}}\, \text{Re}(< \mathbf{t}, \mathbf{r}, \bar{\mathbf{h}} >)$.

Composition and 1-to-N

- **Composition Relations:**

$$r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z) \quad \forall x, y, z$$

  - **Example**: My mother's husband is my father.
- **1-to-N Relations:**
  - **Example**: If (h, r, t_1) and (h, r, t_2) exist in the knowledge graph
- **ComplEx** share the same property with **DistMult**
  - **Cannot model composition relations**
  - **Can model 1-to-N relations**

Today: KG Completion Models

What we learned so far:

Model	Score	Embedding	Sym.	Antisym.	Inv.	Compos.	1-to-N
TransE	$-\|\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✗	✓	✓	✓	✗
TransR	$-\|\|\boldsymbol{M}_r\mathbf{h}+\mathbf{r}-\boldsymbol{M}_r\mathbf{t}\|\|$	$\mathbf{h},\mathbf{t}\in\mathbb{R}^k$, $\mathbf{r}\in\mathbb{R}^d$, $\boldsymbol{M}_r\in\mathbb{R}^{d\times k}$	✓	✓	✓	✓	✓
DistMult	$<\mathbf{h},\mathbf{r},\mathbf{t}>$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{R}^k$	✓	✗	✗	✗	✓
ComplEx	$\mathrm{Re}(<\mathbf{h},\mathbf{r},\bar{\mathbf{t}}>)$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{C}^k$	✓	✓	✓	✗	✓
RotateE	$-\|\|\mathbf{h}\circ\mathbf{r}-t\|\|$	$\mathbf{h},\mathbf{t},\mathbf{r}\in\mathbb{C}^k$	✓	✓	✓	✓	✗

- …Hadamard product:

$$\overset{G}{\begin{bmatrix}3 & 5 & 7\\ 4 & 9 & 8\end{bmatrix}}\circ\overset{H}{\begin{bmatrix}1 & 6 & 3\\ 0 & 2 & 9\end{bmatrix}}=\overset{N}{\begin{bmatrix}3\times1 & 5\times6 & 7\times3\\ 4\times0 & 9\times2 & 8\times9\end{bmatrix}}$$

TransE and RotatE: they both satisfy a weaker notion of 1-to-N - that many tails can be equidistant to r*h / r + h

Sun et al, RotatE: Knowledge Graph Embedding by Relational Rotation in Complex Space, ICLR 2019

KG Embeddings in Practice

1. Different KGs may have **drastically different relation patterns**!
2. There is not a general embedding that works for all KGs, use the **table** to select models
3. Try **TransE** for a quick run if the target KG does not have much symmetric relations
4. Then use more expressive models, e.g., **ComplEx**, **RotatE** (**TransE** in Complex space)

Empirical comparison

Model	FB15k-237			WN18RR		
	MR↓	MRR↑	H10↑	MR↓	MRR↑	H10↑
TransE	357	.294	.465	3384	.226	.501
TransR						
DisMult	254	.241	.419	5110	.43	.49
ComplEx	339	.247	.428	5261	.44	.51
RotatE	**177**	**0.338**	**0.533**	**3340**	**0.476**	**0.571**

Summary of Knowledge Graph

- Link prediction / Graph completion is one of the prominent tasks on knowledge graphs
- Introduce **TransE** / **TransR** / **DistMult** / **ComplEx** models with different embedding space and expressiveness
- **Next:** Reasoning in Knowledge Graphs